

Kizen — Lock In & Learn

Official Problem Statement, Product Specification & Architecture Blueprint

PROJECT CODE	TARGET AUDIENCE	CORE TECH STACK	DESIGN AESTHETIC
KZ-2026-FOCUS	University & Self-Learners	Next.js 14, Tailwind, Firebase	Editorial Light (Default)

1. Executive Summary & Problem Statement

Over 80% of university and self-taught engineering students use YouTube as their primary learning source for computer science, engineering, and exam preparation. However, learners face the critical **YouTube Distraction Paradox**:

- Algorithmic Traps:** Recommendation engines, clickbait thumbnails, intrusive ads, and short-form video feeds derail study sessions within minutes.
- The Academic Curriculum Gap:** University students receive complex course syllabi (e.g., *CSE220: Data Structures*) but waste hours searching YouTube trying to locate high-quality, sequential lectures matching their specific semester outline.
- Passive Consumption:** Students play lectures in background tabs or skim forward to conclude lessons without cognitively retaining information.

The Challenge: Build **Kizen** — an ultra-clean, zero-distraction video learning ecosystem that bridges academic curricula with YouTube's vast knowledge base. Features an autonomous **Syllabus-to-Roadmap Matcher**, a **Proof-of-Focus verification engine**, and daily streak habit tracking.

2. Non-Negotiable Visual Identity Guidelines

CRITICAL FOR DEVELOPERS & AI ASSISTANTS: Strictly avoid generic AI aesthetics. No neon purple/blue gradients, no robotic buzzwords, and zero development jargon visible in user-facing UI copy.

- Default Theme: Light Mode by Default** (Editorial Paper Aesthetic inspired by Notion, Apple, and Linear).
 - Canvas: Pure Paper White (#ffffff) and Soft Ivory (#fafafa).
 - Typography: Deep Ink Charcoal (#18181b) for primary headers, Slate Grey (#64748b) for descriptions.
 - Borders: Sharp, subtle 1px border lines (#e4e4e7) with gentle ambient elevation.
- Toggleable Dark Mode:** Clean deep graphite palette for nighttime learners, accessible via a subtle header switch.
- Zero Cheap AI Clichés:**
 - ❌ NO 4-corner sparkle stars (✨) or ninja star icons.
 - ❌ NO lightning thunderbolt badges (⚡).
 - ❌ NO robotic "AI Magic" tags or tech jargon on user-facing screens.
- Refined Accents:** Forest Emerald (#059669) for verified progress checkmarks and Warm Amber Ember (#d97706) for streak flames.

3. Functional Modules & Requirements

Module A: Flagship X-Factor • Syllabus-to-Roadmap Matcher CORE MOAT

- Syllabus Upload:** Students upload their university course syllabus (PDF) or paste exam topics (e.g., "Unit 1: Linked Lists, Unit 2: Trees, Unit 3: Graphs").

- **Multi-Channel Curation (Not limited to 1 playlist):** The engine extracts topics and searches YouTube across top-rated educational creators to curate the highest-yield video for each specific topic into a cohesive course.
- **Language Preference Selector:** Users select their preferred learning language:
 -  **Bangla** (e.g., Anisul Islam, Learn with Sumit, university teacher lectures)
 -  **English** (e.g., freeCodeCamp, CS50, Abdul Bari, Striver)
 -  **Hindi** (e.g., CodeWithHarry, Apna College, Gate Smashers)
- **Quota-Safe Architecture (Zero Quota Failure):** Uses high-quota Cloudflare AI / rule-based regex parsers for text extraction. YouTube search queries live APIs directly so **no LLM hallucinations or broken video links occur**.

Module B: Smart Playlist Importer & Quota System

- **1-Click YouTube Parser:** Users paste any public or unlisted YouTube playlist URL. The backend parses playlist ID, title, video count, thumbnails, and durations.
- **Custom Renaming:** Users can override the default title with a custom name (e.g., *"30-Day Full-Stack Bootcamp"*).
- **Hard Quota (Max 10 Active Playlists):** Each user is strictly capped at 10 active playlists at any given time. Reaching 10 disables the import button with an intuitive prompt to archive an existing course.
- **Dynamic YouTube Auto-Sync:** If a creator adds new videos to the original YouTube playlist later, Kizen automatically detects and appends them to the end of the queue. All existing verified progress (videos 1..N) and streaks remain 100% intact.

Module C: Cinema Focus Player Engine

- **100% Distraction Stripping:** Complete elimination of YouTube comments, related videos, end-screen popups, like/share overlays, and sidebars.
- **Custom Playback Speed:** Defaults to standard **1.0x**. Supports fine-grained controls: **0.5x, 0.75x, 1.0x, 1.25x, 1.5x, 1.75x, 2.0x, 2.5x, 3.0x**.
- **Strict Playlist Auto-Play:** When a video finishes, **ONLY** the next video in that exact playlist begins playing. No external suggestions can ever load.
- **Embed-Disabled Resilience:** Clean IFrame player with fallback error handling.

Module D: Active Attention & Proof-of-Focus Engine

- **Dynamic Floating Verification Code:** During playback, a unique 4-to-6 digit numeric code floats across a random coordinate on the player for 15–20 seconds at an unpredictable timestamp (between 30% and 80% of video progress). The user notes this code.
- **Watch-Time Gate (Anti-Cheat):** Scrubbing or fast-forwarding to the end is blocked from completing. The submission box remains locked until the user has genuinely watched at least **80% of total video duration**.
- **End-of-Video Submission:** Once the 80%+ threshold is reached and the video ends, the code submission box unlocks. Submitting the correct code awards **+50 XP**, applies a permanent verified checkmark **[✓]**, and increments the Daily Streak.
- **Micro-Video Exemption:** Videos under 5 minutes do not require code entry; completing 90% watch time automatically grants verified status.

Module E: In-House Focus Trap & UX Balance

- **Zero External Extension Requirement:** All focus enforcement runs natively within the web application.
- **Window Blur & Focus Trap:** Page Visibility API detects when the user switches tabs or focuses on external windows.
- **3–5s Grace Period:** Prevents annoying users who accidentally click outside. Continued absence triggers a gentle pause and warning screen.
- **Two Dedicated Study Modes:**
 - *Lecture Mode:* Strict window-blur lockdown for conceptual lectures.
 - *Studio / Code-Along Mode:* Allows split-screen IDE or terminal usage while maintaining the 80% watch-time gate and floating code requirement.

Module F: Gamification, Streaks & Note-Taking

- **Daily Streak Counter (🔥):** Increments with at least one verified video completed per calendar day. Timezone-aware midnight reset.
- **365-Day Activity Heatmap:** GitHub-style yearly contribution grid displaying study volume per day.
- **Personal Best & Consistency Tracking:** Focuses on self-growth metrics ("You studied 3 more days than last week!").

- **Timestamped Notes:** Side-by-side markdown notepad with a `[+ Timestamp]` button (e.g., `[04:25]`) linking directly to specific video moments.

4. System Architecture & Technical Stack

Layer	Technology	Architectural Purpose
Frontend	Next.js 14+ (App Router, TypeScript)	Server & Client components, fast routing, responsive desktop layouts.
Styling & Icons	Tailwind CSS + Lucide Icons	Editorial light theme, crisp typography, zero bloated CSS.
Database & Auth	Google Firebase (Cloud Firestore + Auth)	100% persistent cloud storage, zero inactivity sleep/pauses, Google OAuth & Email auth.
Curriculum Engine	Cloudflare AI / Regex NLP + YouTube API	Parses syllabus topics and executes live searches with language filtering.
Mobile Ready	Flutter SDK Integration Ready	Same Firebase schema and authentication tokens directly map to future Flutter mobile apps.

5. Database Schema & Data Models (Cloud Firestore)

```
// Collection: users
interface UserProfile {
  uid: string;
  email: string;
  displayName: string;
  photoURL?: string;
  totalXP: number;
  currentStreak: number;
  lastStudyDate: string; // "YYYY-MM-DD"
  activePlaylistsCount: number; // Max 10
  createdAt: string;
}

// Collection: playlists
interface Playlist {
  id: string;
  userId: string;
  ytPlaylistId?: string; // Optional if AI syllabus roadmap
  customTitle: string;
  thumbnailUrl: string;
  totalVideos: number;
  completedVideos: number;
  sourceType: "youtube_playlist" | "syllabus_roadmap";
  languagePreference?: "bn" | "en" | "hi";
  lastSyncedAt: string;
  createdAt: string;
}

// Collection: video_progress
interface VideoProgress {
  id: string; // `${userId}_${ytVideoId}`
  userId: string;
  playlistId: string;
  ytVideoId: string;
  videoTitle: string;
  durationSeconds: number;
  watchedSeconds: number;
  isCompleted: boolean;
  isVerified: boolean;
  completedAt?: string;
}

// Collection: daily_activity (For 365-Day Heatmap)
interface DailyActivity {
  id: string; // `${userId}_${dateString}`
  userId: string;
  date: string; // "YYYY-MM-DD"
  watchSeconds: number;
  verifiedVideosCount: number;
  xpEarned: number;
}
```

6. Instructions for AI Coding Assistants (Cursor / Claude / Copilot Prompt)

HOW TO USE THIS WITH AN AI AGENT: Pass this prompt directly to your AI tool to scaffold and build Kizen systematically.

You are tasked with building "Kizen – Lock In & Learn", a distraction-free, gamified YouTube learning platform.

Follow these strict constraints:

1. DESIGN: Implement a default Editorial Light Mode (Paper White #ffffff, Soft Ivory #fafafa, Deep Charcoal #18181b, Emerald #059669, Amber #d97706). DO NOT use AI cliches (no purple gradients, no sparkle star icons, no technical jargon in UI). Provide an optional Dark Mode.
2. SYLLABUS MATCHER (FLAGSHIP): Allow users to upload or paste a course syllabus/exam topics. Extract module topics and query YouTube to compile a curated roadmap in the user's chosen language (Bangla, English, or Hindi).
3. PLAYER: Embed YouTube videos via YouTube IFrame API. Hide all related videos, comments, and recommendations. Support playback speeds from 0.5x to 3.0x (default 1.0x).
4. PROOF-OF-FOCUS: Display a floating 4-digit verification code at a random timestamp (between 30% and 80%). Lock verification submission until the user has genuinely watched at least 80% of the video. Submit code at the end of the video to grant XP and mark verified [✓].
5. PLAYLISTS: Allow users to paste a YouTube playlist URL. Automatically fetch metadata. Enforce a strict maximum limit of 10 active playlists per user. Automatically sync newly added videos when the user visits, keeping existing progress intact.
6. FOCUS TRAP: Detect window blur and tab switching. Include a 3-second grace period. If focus is not regained, auto-pause the video and display a clean focus-lost overlay. Support Lecture Mode and Studio/Code-Along Mode.
7. GAMIFICATION: Track daily streaks (🔥), total XP, a 365-day GitHub-style contribution heatmap, and self-consistency metrics.
8. PERSISTENCE: Use Firebase Auth (Google & Email) and Cloud Firestore. Ensure schema is ready for future Flutter mobile app consumption.

7. Hackathon Evaluation & Scoring Rubric

Criteria	Weight	Evaluation Benchmark
Curriculum Problem Solving	25%	Effectiveness of Syllabus-to-Roadmap matching and multi-language YouTube curation.
Proof-of-Focus Engine	25%	Reliability of random floating code generation, 80% watch gate, and anti-cheat validation.
UI/UX Craftsmanship	20%	Clean editorial light mode, zero AI clichés, responsive cinema layout, smooth interactions.
Gamification & Retention	15%	Accurate streak calculation, heatmap data rendering, and personal-best consistency.
Architecture & Code Quality	15%	Clean Next.js App Router code, Firebase persistence, and Flutter-ready data models.